STUMPRIDER · TRACEABILITY MATRIX

StumpRider traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
Source	docs/03b-traceability.md @ 6afbf98		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

StumpRider traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 11 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Free a snagged net	SMR-CAL-001, SMR-BLD-001, SMR-STM-001, SMR-OPT-001	Analysis (SMR-CAL-001 s4), TRL 3; staged-snag trials with the hook head beside the ring, TRL 4	At risk: hooked 81 to 126 N, inside the pin [F1, F2]; wrapped 368 N to 2.3 kN [F3, F4], 162 to 493 N once worked back half a turn [F7, F8]
UR-002	PR-002	Reach	SMR-CAL-001, SMR-BLD-001	Analysis (SMR-CAL-001 s5), TRL 3; measured at the trial site, TRL 4	Met on paper: 4.18 m [R1]; tether 8 m [R3]
UR-003	PR-003	Limit the side pull	SMR-CAL-001, SMR-BLD-001, SMR-STM-001	Analysis (SMR-CAL-001 s3), TRL 3; CalRig pull tests and heel tests by canoe class, TRL 4	At risk: static heel 3.4 deg kneeling, 4.4 deg standing [H5, H6] on screening hydrostatics (5.1 deg with a 1.0 m beam [Q1]); 6.5 deg peak for a quick pull held just under the pin [H11], ruled out by the steady-lift rule and measured at TRL 4 (O6, decided 2026-10-25); measured righting-moment criterion added (O9, decided 2026-10-25)
UR-004	PR-004	Protect the net	SMR-CAL-001	Mesh count after staged releases, TRL 4	To verify at TRL 4: the pin caps the pull at 347 N [P3]
UR-005	PR-005	Fast to use	SMR-CAL-001	Timed trials with fisher crews, TRL 4	To verify at TRL 4
UR-006	PR-006	Fit any net line	SMR-CAL-001, SMR-BLD-001	Analysis and inspection (SMR-CAL-001 s5), TRL 3; fit check on sample lines, TRL 4	Met on paper: hinged ring, 100 mm inside [R4]
UR-007	PR-007	Stows in a canoe	SMR-CAL-001, SMR-BLD-001, SMR-STM-001	Analysis (SMR-CAL-001 s8), TRL 3; weighed at TRL 4	At risk: met on paper on the bamboo local variant, 3.95 kg [V3] (decision 2A, 2026-10-04), but the 0.05 kg margin after F1 is inside the uncertainty (wet bamboo, zinc, paint; independent review); aluminium prototype 4.34 kg [M11] recorded, not a failure. Packed 1.55 m [M9]
UR-008	PR-008	Survive wet storage	SMR-CAL-001, SMR-	Analysis (SMR-STM-001 S11 to S13), TRL 3; field	Met on paper: galvanised steel, 6063 aluminium, stainless pins and rivets, polyester, polyethylene; treated bamboo

User requirement	Product requirement	Title	Evidence	Verification	Status
			BLD-001, SMR-STM-001	durability check with partner crews, TRL 4	and zinc-painted ferrules in the local variant to watch
UR-009	PR-009	Locally buildable	SMR-CAL-001	Analysis (SMR-CAL-001 s9), TRL 3; build trial with a local smith, TRL 4	At risk: about 8.1 h [T1], galvanising sent out
UR-010	PR-010	Affordable	SMR-CAL-001	Costed BOM (SMR-CAL-001 s9, s10), TRL 3; local prices, TRL 4	Not met: USD 66.90 bamboo local variant [V8] (USD 54.90 before O8); USD 95.50 aluminium prototype [V9]
UR-003	PR-011	Calibrated pin wire	SMR-CAL-001, SMR-BLD-001, SMR-STM-001	Inspection of the reel and CalRig breaks (SMR-BLD-001 first checks), TRL 4	To verify at TRL 4 (SMR-STM-001 S1, S16)
UR-003	PR-012	The pin is the weakest link	SMR-CAL-001, SMR-STM-001	Analysis (SMR-CAL-001 s6), TRL 3	At risk: aluminium prototype met on paper (smallest factor 2.21, pole buckling [S2]; hook head 3.1 after F1 [S12]; pole side bending 2.4 [S17]); hook head with the bight on the tip leg 1.58 at the bend [S13] and 1.87 in the straight shank [S13a] (O10: the approved gusset was stopped on 2026-10-25 because it does not reach the shank, back with Amish); bamboo pole buckling 1.49 [V5]
UR-011	PR-013	Marked and handed out with protection work	SMR-CAL-001, SMR-BLD-001	Inspection (BOM line 19; SMR-BLD-001 safety stops), TRL 3; partner distribution plan, TRL 4	Met by design for the rating mark; tag artwork and distribution plan to do with the partner

Gaps

None.